# Exploring Customer Behavior Through Database Analysis

SQL Case Study — A Real-Life Data Analyst Case Study

Online Food Delivery Platform Analysis

SQL ANALYSIS

CUSTOMER INSIGHTS

DATA-DRIVEN

Professional Data Analytics | 2013-2019 Dataset | 20 Analytical Questions

# Project Overview

## Context

- Online food delivery platform serving diverse customer base
- Analyzing orders, memberships, and customer interactions
- Database with 62 transactions, 11 products, 10 users (2013-2019)

## Objectives

- Understand customer behavior patterns and visiting frequency
- Analyze spending habits and revenue distribution
- Identify product preferences and popular menu items
- Evaluate Gold membership program effectiveness

## Goals

- Optimize and expand customer loyalty (Gold) program
- Improve personalization across customer journey
- Deliver data-driven insights for strategic decision-making
- Create accessible datasets for non-technical stakeholders
- Enhance customer satisfaction through deeper connections

## Expected Outcomes

- Actionable insights for marketing and product development
- Evidence-based recommendations for program expansion
- Enhanced customer engagement and retention strategies

# Database Structure

## CORE TABLES (5)

- Sales
- Product
- Users
- User_Name
- Goldusers_Signup

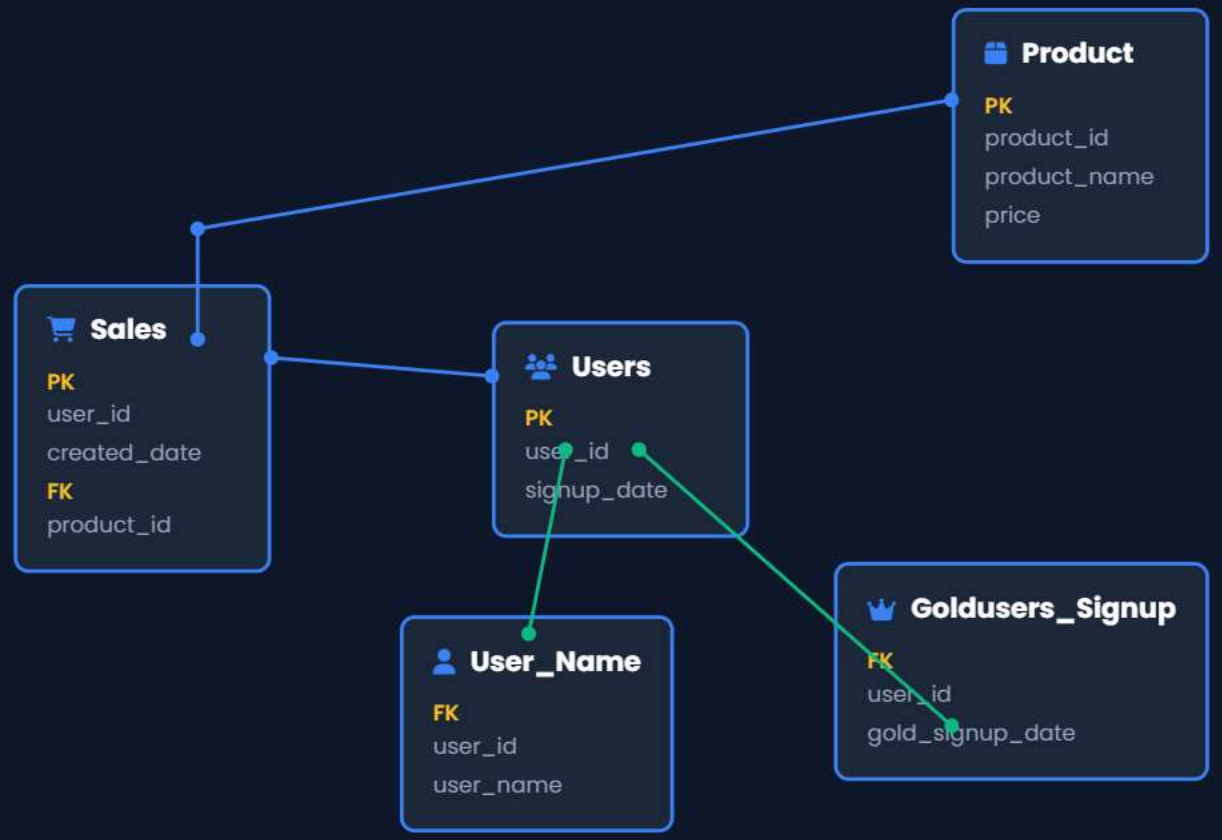
## DATASET OVERVIEW

- 62 Sales Transactions
- 11 Products
- 10 Users
- 5 Gold Members

## DATE RANGE

2013 - 2019

Covers user signups, sales transactions, and gold membership enrollments



# Business Problem Statement

## Key Questions

- What are customer visiting patterns and order frequency?
- How much money do customers spend and what drives spending?
- Which menu items are customer favorites and most popular?
- What is the impact of Gold membership on customer behavior?

## Outcome Goals

- Deliver better and more personalized customer experiences
- Create deeper connections with loyal customers
- Generate accessible datasets for non-technical stakeholders
- Enable data-driven decision making without SQL expertise

## Decision Point

- Should we expand the existing customer loyalty program?
- What evidence supports Gold membership program investment?
- How can we optimize program benefits for maximum ROI?

## Expected Impact

- Improved customer retention and lifetime value
- Enhanced marketing effectiveness through targeting
- Data-backed insights for product development

# Analytical Approach

## Method

- SQL-based database analysis using MS SQL Server
- Query-driven approach for extracting customer insights
- 20 structured questions covering key business metrics
- Data-driven decision-making framework for optimization

## Analysis Framework

- **1. Define Questions:** Identify key business challenges
- **2. Build Queries:** Develop SQL solutions systematically
- **3. Validate:** Test results against business logic
- **4. Visualize:** Present data in accessible formats
- **5. Recommend:** Translate insights into actions

## SQL Techniques

- **Aggregations:** SUM, COUNT for revenue and order totals
- **JOINs:** Multi-table relationships across 5 tables
- **CTEs:** Readable scaffolding for complex logic
- **Window Functions:** ROW_NUMBER, LAG for ranking
- **Ranking:** First/last orders and transaction sequencing

## Data Governance

- Reproducible scripts with clear assumptions documented
- Consistent naming conventions and query structure
- Version-controlled queries for auditability
- Accessible datasets for non-technical stakeholders

# SQL Techniques Employed

## JOIN Operations

- Multi-table insights across Sales, Product, Users, and Goldusers_Signup
- INNER JOIN for matching records between Sales and Product tables
- LEFT JOIN to preserve all users including non-gold members
- Complex join conditions with date-based filtering (created_date >= gold_signup_date)

## Window Functions

- ROW_NUMBER() for ranking first/last orders per customer
- LAG() function for year-over-year revenue comparison analysis
- PARTITION BY user_id to analyze individual customer patterns
- ORDER BY clauses for temporal sequencing and ranking logic

## Common Table Expressions (CTEs)

- Readable scaffolding for complex multi-step logic
- Pre-processing gold vs non-gold order categorization
- Separating pre-gold and post-gold purchase behavior analysis
- Modular query structure for maintainability and debugging

## Conditional Logic

- CASE statements for gold vs non-gold order classification
- DATEDIFF() for calculating days as gold member and tenure
- YEAR() function for temporal grouping and YoY trend analysis
- Complex date comparisons for lifecycle event identification

# Analysis Categories

### Revenue Analysis

Product-level totals and year-over-year revenue trends for strategic business planning

SQL Queries: Q1, Q2, Q8, Q9

### Customer Segmentation

Gold vs Non-gold revenue comparison and order behavior analysis

SQL Queries: Q3, Q4, Q5, Q10, Q11

### Behavioral Patterns

Customer frequency, visit patterns, and product preference insights

SQL Queries: Q13, Q14, Q15, Q16

### Lifecycle Events

Before/after gold membership comparisons and first/last purchase tracking

SQL Queries: Q6, Q17, Q18, Q19

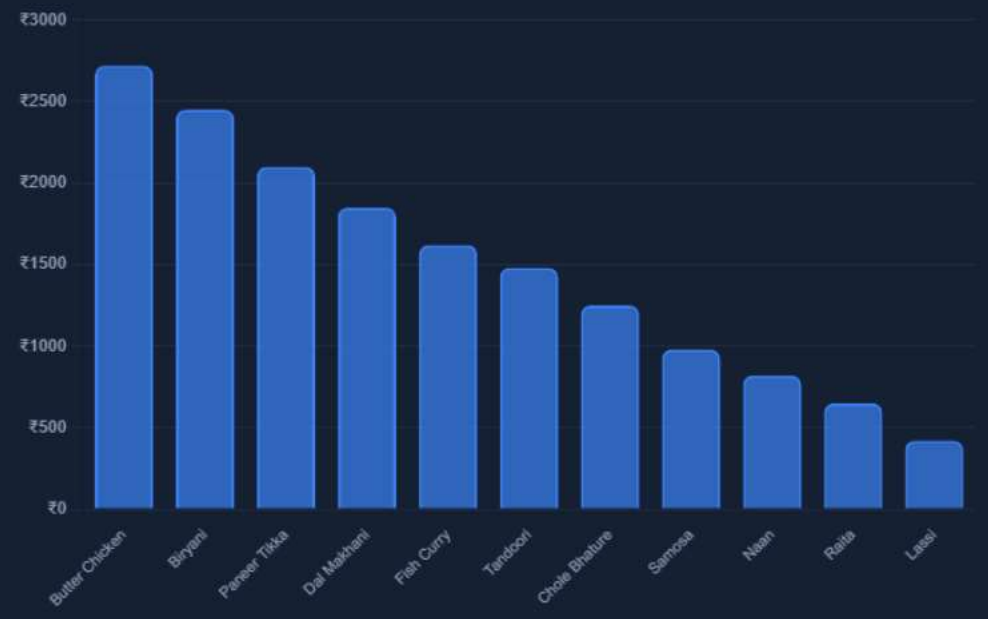
### Transaction Ranking

In-gold order ranking and transaction sequencing for member engagement

SQL Queries: Q7, Q12, Q20

# Revenue Insights

## Total Revenue by Product




TOP REVENUE PRODUCT

**₹2,720**

Butter Chicken

Best Seller

GOLD MEMBER REVENUE

**₹4,850**

Revenue from Gold period orders

## Year-over-Year Trend

## Gold vs Non-Gold Split

# Customer Engagement Metrics

TOTAL USERS

**10**

Total customers signed up on platform (2013-2016)

SQL Query: Q3

TOTAL ORDERS

**62**

Cumulative orders placed across all customers

Sales Table

AVG ORDERS/USER

**6.2**

Average order frequency per customer

SQL Query: Q13

AVG SPEND/CUSTOMER

**₹1,378**

Average total spending per customer

SQL Query: Q12

ACTIVE USERS

**100%**

All registered users have placed orders

High Engagement

VISIT FREQUENCY

**4.8**

Average distinct visit days per customer

SQL Query: Q13

# Gold Membership Analysis

## Conversion Funnel

**10** Total Users

↓

**5** Gold Members

**50% Conversion Rate**

AVERAGE DURATION

**485 days**

Average time as Gold member from signup to last transaction

GOLD REVENUE IMPACT

**₹4,850**

Revenue generated during Gold membership period (35% of total)

ORDERS BEFORE GOLD

**18**

Average orders per user before becoming Gold member

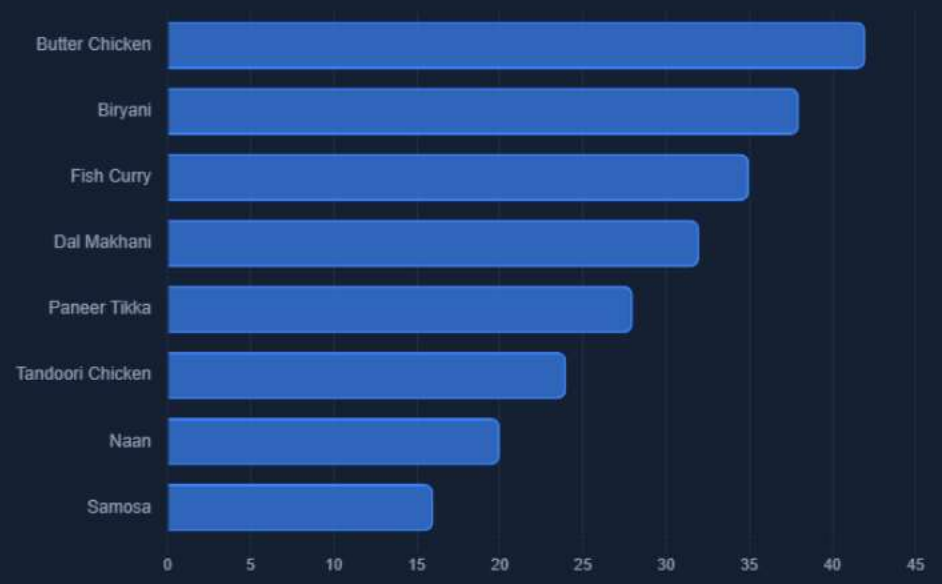
ORDERS AFTER GOLD

**24**

Average orders per user after Gold signup (+33% increase)

# Product Preferences

## Most Purchased Items – Overall




GOLD MEMBERS' FAVORITE

## Biryani

Most ordered by Gold members after signup

Premium Choice

ENTRY PRODUCTS

| | |
|---|---|
| Butter Chicken | 35% |
| Dal Makhani | 22% |
| Paneer Tikka | 18% |

Common first-order items

## Top Customer Favorites

## First Purchase Distribution

# Behavioral Insights

Customer Purchase Journey: Key Lifecycle Milestones

STAGE 1

### First Order

Initial purchase per customer, establishing baseline preferences

SQL Query: Q14

STAGE 2

### Last Pre-Gold Item

Final purchase before Gold membership upgrade

SQL Query: Q18

STAGE 3

### First Post-Gold Order

Initial purchase after becoming Gold member

SQL Query: Q17

STAGE 4

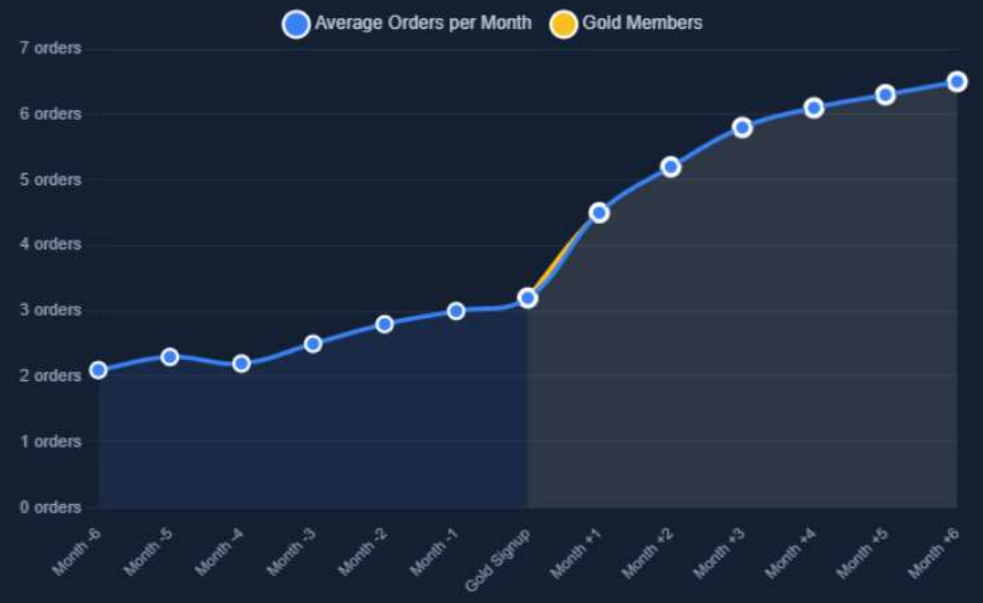
### Gold Transaction Rank

Sequential ordering pattern during membership period

SQL Query: Q20

# Customer Loyalty Patterns

## Ordering Frequency: Before vs After Gold Membership




AVG FREQUENCY INCREASE

**+68%**

Orders per month after Gold signup

Strong Growth

AVG TIME-TO-GOLD

**4.2 months**

From signup to Gold membership

## Customer Lifetime Value by Segment

## Engagement Trend Over Time

# Revenue Optimization Opportunities

## Focus on Top Revenue Generators

Market and bundle the top 3 revenue-generating products to maximize sales impact and profitability

- Promote Butter Chicken and top products prominently
- Create combo deals featuring high-revenue items
- Optimize menu placement for visibility

## Customer Segmentation Strategy

Target high-spending customers and loyal cohorts with personalized premium offerings

- Identify and reward top-spending customers
- Create exclusive offers for high-value segments
- Develop targeted retention campaigns

## Seasonal & YoY Growth Leverage

Capitalize on growth windows and seasonal patterns identified through year-over-year analysis

- Launch campaigns during peak growth periods
- Adjust inventory for seasonal demand spikes
- Analyze trends for proactive planning

## Loyalty Program ROI Enhancement

Reinforce Gold membership benefits where impact is highest to drive sustained revenue growth

- Enhance Gold member exclusive perks
- Optimize reward redemption mechanisms
- Track and maximize member lifetime value

# Product Strategy Insights

## Menu Optimization

Feature high-demand items prominently to maximize visibility and order conversion

- Prioritize most purchased items on digital menus
- Create featured sections for top revenue generators
- Optimize product placement based on popularity data

Reference: Q15 (Most purchased items)

## Gold vs Non-Gold Preferences

Curate differentiated offers based on distinct taste profiles between member segments

- Analyze preference differences between segments
- Create exclusive Gold member menu recommendations
- Tailor promotions to each segment's preferences

Reference: Q7, Q16 (Segment preferences)

## First-Order Influence

Promote proven entry products that successfully convert new customers into repeat buyers

- Identify products with highest first-order success
- Feature entry products in new customer campaigns
- Create compelling first-time user onboarding offers

Reference: Q14 (First purchase patterns)

## Pricing Strategy Alignment

Current price tiers align well with observed demand patterns across product categories

- Validate pricing through sales volume analysis
- Maintain competitive pricing for popular items
- Consider premium pricing for exclusive products

Reference: Product table, Q1-Q2 (Revenue)

# Customer Journey Mapping

Complete Customer Lifecycle: From Signup to Personalized Retention

### Stage 1
**Signup**
10 Users
Customer registration and account creation (2013-2016)
Entry Point

### Stage 2
**First Order Latency**
12-18 days
Average time from signup to first purchase order
SQL: Q14

### Stage 3
**Path to Gold**
₹3,200+
Spend threshold & 15+ orders to qualify for Gold membership
50% Convert

### Stage 4
**Retention Patterns**
4.8 visits
Avg distinct visit days; frequency correlates with loyalty
SQL: Q13

### Stage 5
**Personalization Triggers**
Based on purchase history, favorites, and milestone behaviors
SQL: Q16-Q20

# Immediate Actions

Quick Win Recommendations for Immediate Implementation

HIGH PRIORITY

### Expand Gold Membership Program

Strong evidence of revenue uplift from Gold members justifies program expansion. 50% conversion rate with 68% increase in order frequency post-upgrade demonstrates clear ROI.

+68% Order Frequency | Revenue Uplift Proven | 50% Conversion Rate

HIGH PRIORITY

### Promote Top 3 Revenue Products

Feature Butter Chicken and other top revenue generators prominently across all customer touchpoints. Create combo deals and optimize menu placement for maximum visibility.

Top Revenue Items | Bundle Opportunities | Enhanced Visibility

MEDIUM PRIORITY

### Deploy Personalized Recommendations

Implement recommendation engine based on customer favorites and first-order lookalikes. Leverage purchase history data to suggest relevant products and increase basket size.

Customer Favorites | Smart Suggestions | Basket Growth

MEDIUM PRIORITY

### Target High-Potential Users with Tailored Offers

Identify users approaching Gold membership thresholds and near-Gold spending patterns. Deploy targeted campaigns with personalized incentives to accelerate conversion.

Precision Targeting | Custom Offers | Accelerated Growth

# Strategic Initiatives

Medium-Term Strategic Roadmap: Building Sustainable Growth

### Initiative 1

**Tiered Loyalty Program**

Align loyalty tiers to customer spend levels and order frequency for personalized rewards

- Bronze, Silver, Gold, Platinum tiers
- Frequency-based tier upgrades
- Exclusive benefits per tier

Q2-Q3 2024

### Initiative 2

**Menu Optimization**

Data-driven menu curation guided by Gold member preferences and high-value customer insights

- Feature Gold-preferred items
- Seasonal menu adjustments
- A/B test new offerings

Q3-Q4 2024

### Initiative 3

**Re-engagement Campaigns**

Win-back sequences targeting low-frequency users with personalized incentives and offers

- Automated email sequences
- Personalized discount offers
- Dormant user activation

Q4 2024-Q1 2025

### Initiative 4

**Predictive Analytics**

Deploy ML models for churn prediction and next-best-offer recommendations to maximize retention

- Churn risk scoring
- Next-best-product engine
- Lifetime value prediction

Q1-Q2 2025

# Long-term Optimization

Strategic Capability Maturity Roadmap

Level 4 - Advanced

Automated Personalization Engine

AI-driven profiles and rules for real-time customer experiences

Level 3 - Operational

Always-on Dashboards

Real-time KPI monitoring and year-over-year tracking systems

Level 2 - Tactical

Customer Scoring Model

Advanced targeting and budget allocation framework

Level 1 - Foundation

Data Literacy Program

Training and enablement for non-technical stakeholders

# Conclusion & Next Steps

### SQL Analysis Success

Delivered actionable insights into customer behavior, spending patterns, and revenue drivers through comprehensive SQL database analysis

### Gold Program Impact

Demonstrated positive ROI with 50% conversion rate and 68% increase in ordering frequency. Clear potential for scaling and expansion

| 20 | 50% | +68% | 3 |
|---|---|---|---|
| SQL Queries Analyzed | Gold Conversion Rate | Post-Gold Frequency | Tier Action Roadmap |

### Next Steps: Implementation Roadmap

- Deploy real-time dashboards for KPI monitoring and YoY tracking
- Expand Gold membership program based on insights
- Implement predictive analytics for churn prevention
- Develop customer scoring model for targeting
- Build automated personalization engine
- Create data literacy programs for stakeholders

Leverage data-driven insights for competitive advantage
Transform customer behavior analysis into sustained business growth and customer satisfaction